

Trainings ▾

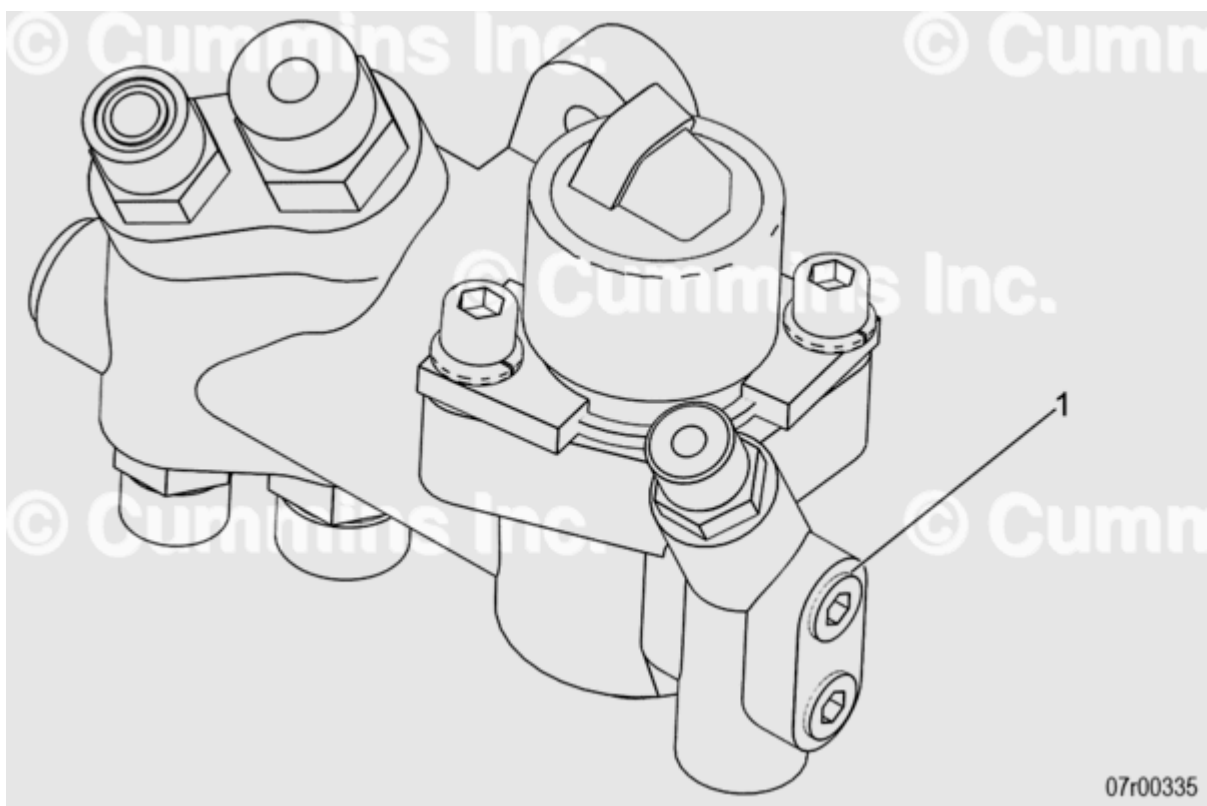
General Information

The K38, K50, QSK38 and QSK50 liter engines have an optional hydraulic fan clutch available.

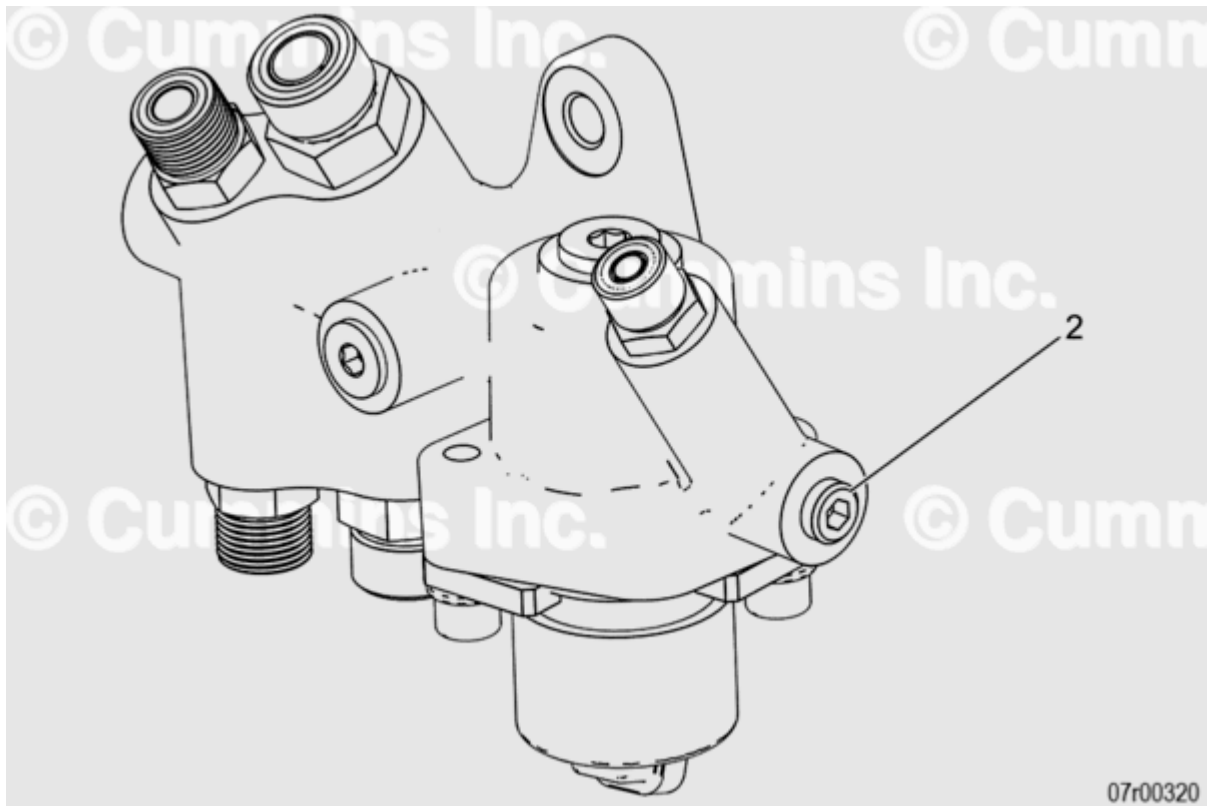
The hydraulic fan clutch uses engine oil from the main oil rifle in the engine block, through a control solenoid and distribution manifold, to actuate the fan clutch hub.

There are two different fan clutch lubricating oil manifolds. The manifolds are identified by the difference in solenoid mounting location. The solenoid will be mounted either facing upwards (1) or downwards (2). All service procedures remain the same.

Clutch solenoid replacement does **not** require the lubricating oil manifold to be removed. Solenoid-**only** replacement can be carried out on engine.



Solenoid Mounting Location (Upward)



Solenoid Mounting Location (Downward)

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- If the fan clutch lubricating oil manifold is being replaced, drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)

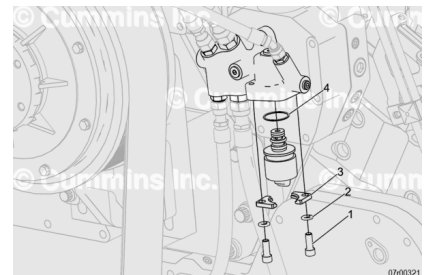
Remove

Lubricating Oil Solenoid

Disconnect the solenoid connector.

Loosen and remove the solenoid mounting capscrews (1), washers (2) and solenoid retainers (3).

Remove the solenoid from its mounting location. Discard the O-ring (4).



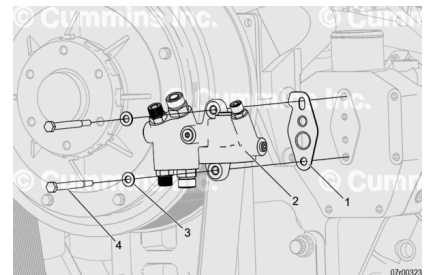
Clutch Manifold

Disconnect the five oil hoses and cover the hose fittings.

Loosen the unions and plugs.

Loosen the two mounting capscrews (4) and washers (3), then remove the manifold assembly (2).

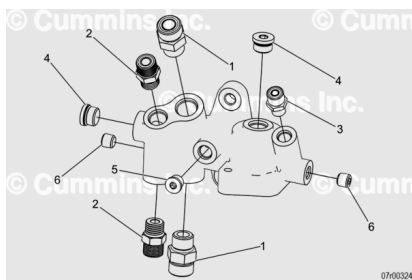
Discard the gasket (1).



Disassemble

Clutch Manifold

Remove all unions and plugs.



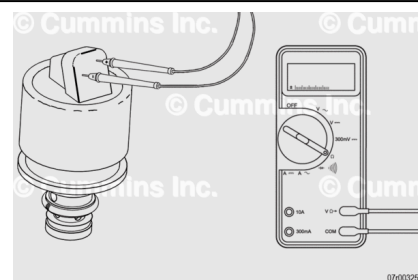
Clean and Inspect for Reuse

Lubricating Oil Solenoid

Use a multimeter to check the solenoid resistance value.

If the resistance value across the solenoid is outside of 52-55 Ohms at 20°C [68°F], the solenoid **must** be replaced.

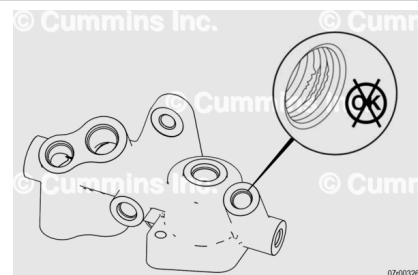
Wipe clean the solenoid o-ring groove on the oil manifold.



Clutch Manifold

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

With all plugs removed, use solvent to clean the manifold.

Use compressed air to clear manifold oil galleries.

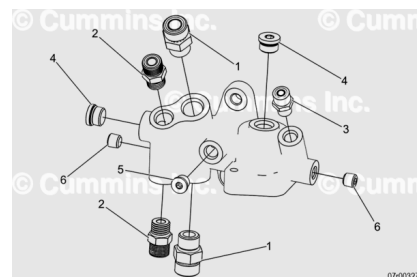
Check the manifold for thread damage, cracks, and wear.

Assemble

Clutch Manifold

Install all unions and plugs. Fit hand tight.

1. 7/8" unions
2. 3/4" unions
3. 9/16" union
4. 3/4" plugs
5. 9/16" plug
6. 1/4" plugs



Install

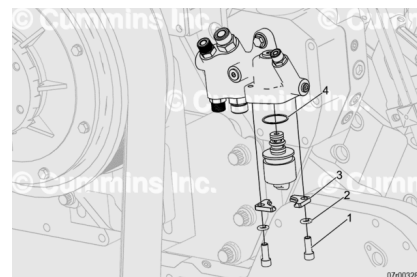
Lubricating Oil Solenoid

Install a new solenoid O-ring (4).

Position the solenoid in the manifold. Align the solenoid connector to the 8 O'clock position when viewed from the underside when fitted.

Install the retainers (3), washers (2) and capscrews (1).

Snug the capscrews by hand.



Clutch Manifold

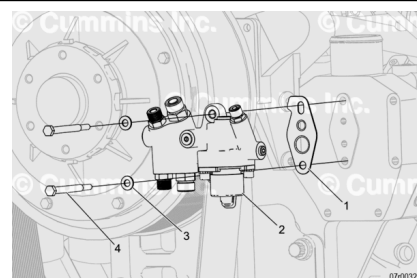
Align the manifold mounting capscrews (4), washers (3) and single piece gasket (1) with the manifold (2). The gasket **must** be orientated as shown.

Snug the bolts by hand then tighten.

Torque Value: 37 n•m [27 ft-lb]

Torque the solenoid mounting capscrews.

Torque Value: 47 n•m [35 ft-lb]



Torque each union and plug.

Torque Value:

7/8" unions (1) 68 n•m [50 ft-lb]

Torque Value:

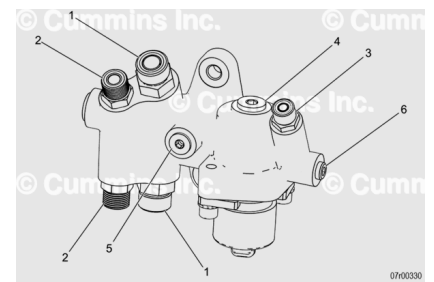
3/4" unions (2) 45 n•m [33 ft-lb]

Torque Value:

9/16" union (3) 37 n•m [27 ft-lb]

Torque Value:

9/16" plug (5) 27 n•m [240 in-lb]



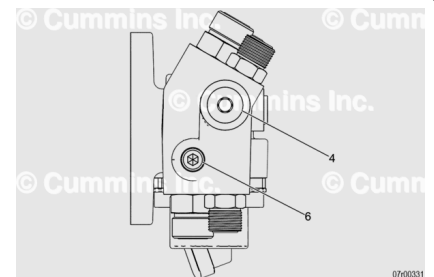
Torque each union and plug.

Torque Value:

3/4" plugs (4) 40 n•m [30 ft-lb]

Torque Value:

1/4" plugs (6) 11 n•m [96 in-lb]



Uncover and connect the five oil hoses as per the original installation. Torque the oil hoses.

Torque Value:

7/8" solenoid drain hose (1) 65 n•m [48 ft-lb]

Torque Value:

45-degree oil control hose (2) 26 n•m [228 in-lb]

Torque Value:

Clutch oil feed hose (3) 45 n•m [33 ft-lb]

Torque Value:

Manifold oil feed hose (4) 54 n•m [40 ft-lb]

Connect the solenoid wiring harness to the solenoid plug.

Note : Installing the downward solenoid type manifold, where the upward solenoid type manifold was previously installed, will require re-routing the solenoid harness. Using two P-clips (5), the harness must follow the gear cover bolts shown to verify correct fitment.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- If the fan clutch oil manifold has been replaced, Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.